Verification Report

For Isometric

CDR Supplier	Charm Industrial
CDR Pathway	Bio-oil Geological Storage
Project	Charm Bio-oil Geologic Storage
Location	Bio-oil Production: AE Cote-Nord Canada Bioenergy Inc (AECN) @ Produits Forestiers Arbec S.E.N.C., 175 Bd du Portage des Mousses Port-Cartier, QC G0W 1X0, Canada 50.025323, -66.818910 Charm Industrial Miniforge 2360 S Rollie Ave Fort Lupton, CO, USA 80621 40.076570, -104.804680 Kerry, Inc. Red Arrow 4343 Expo Drive Manitowoc, WI 54220 44.080613, -87.704757 Bio-oil Injection & Storage: Charm Industrial (Basco Injection site) Evangeline Parish, LA 30.916251, -92.297908
Verifiers	Lily Schacht Tim Hansen
Crediting Period	March 1, 2024 – February 28, 2029
Reporting Period	July 1 – 31, 2026
Net CO_2e Removed	874.29 tonnes CO2e
Report Date	August 14, 2026
Version	V1.0

350 SOLUTIONS

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Acronyms and Chemical Definitions

AECN	AE Cote-Nord Canada Bioenergy Inc
BCU	Book and Claim Unit
BOL	Bill of Lading
COA	Certificate of Analysis
dMRV	Digital Monitoring, Reporting, and Verification
FAR	Forward Action Request
GHG	Greenhouse Gas
KDHE	Kansas Department of Health & Environment
LDENR	Louisiana Department of Energy & Natural Resources
PDD	Project Design Document
REC	Renewable Energy Certificate
RP	Reporting Period
UIC	Underground Injection Control

ISOMETRIC VERIFICATION REPORT		
Company: Charm Industrial	Company Contact: Max Lavine Garrett Lutz*	VVB: 350Solutions Tim Hansen Lily Schacht
Removal Method: Bio-oil Geological Storage		
Document No: 350-IS.CHA.BO.18-VR		

**primary contact/lead author*

1. Introduction

350Solutions was contracted to verify Charm Industrial's greenhouse gas (GHG) statement for the Charm Bio-oil Geologic Storage for the reporting period July 1 – 31, 2026. The validation report for the Charm Kansas Bio-oil Geologic Storage project was completed on June 25, 2024, by Futurepast, which focused on the use of third-party produced bio-oils and was issued by Isometric for the Crediting Period March 1, 2024 – February 28, 2029. An additional validation of the Charm Bio-oil Geologic Storage project for use of bio-oil produced by Charm at the Fort Lupton Miniforge facility was completed on October 18, 2024, for the same crediting period. These validations remain valid for this verification reporting period. Further details and findings can be found in the validation report.[1] Validation of an additional injection and storage location in Evangeline Parish, Louisiana was completed in September 2025 and is the injection and storage site for all injections during this reporting period. This storage site validation is included in the August 1 – 15, 2025 verification report.[2] Validation of a new feedstock used by Charm was completed in December 2025, and a new feedstock for bio-oil produced by Kerry, Inc. for injection was completed on February 12, 2026.

The verified carbon removal for the current reporting period is 874.29 tonnes CO2e.

The verification was performed in accordance with the applicable Isometric Standard, protocol, and modules (Table 1). This verification involved a desk review of the GHG statement and associated evidence, including all social and environmental safeguarding requirements.

A summary of the project, GHG statement and verification is provided in Table 1.

[1] https://registry.isometric.com/project/prj_1HZSSWBQM1S08H83#project-documentation

[2] https://storage.googleapis.com/isometric--production--eu--public-documents/spl_1H31MW3T81S0R67B/prj_1HZSSWBQM1S08H83/pjd_1K5XPX61T1S0TM0Z/verification_report.pdf

Table 1. Verification Summary

GHG Removal Supplier	Charm Industrial
Project Name	Charm Bio-oil Geologic Storage
Project Code	7BDE
Removal Method	Bio-oil Geological Storage
Project Proponents	CDR Supplier: Charm Industrial Feedstock Supplier: Produits Forestiers Arbec S.E.N.C (Quebec mill waste supplier); Altitude Forestry (CO forestry fire prevention waste supplier); Kerry Inc. Geologic Storage Management: Charm Industrial Salt Cavern Storage Management (Prior Injections): Vaulted Deep Regulatory Agency (Injections / Storage): Louisiana Department of Energy & Natural Resources; Kansas Department of Health & Environment (KDHE) (Prior Injections)
Legal Ownership of Removal Credits	Charm Industrial
Validation Report Date	June 25, 2024
Verification Report Date	August 14, 2026
Crediting Period	March 1, 2024 – February 28, 2029
Reporting Period	July 1 – 31, 2026
GHG Statement ID	ggs_1KZBX6AX41S029JS
Net CO_2e Removed	874.29 tonnes CO_2e
Project Location (Address and GPS Coordinates)	<u>Bio-oil Production:</u> AE Cote-Nord Canada Bioenergy Inc (AECN) @ Produits Forestiers Arbec S.E.N.C., 175 Bd du Portage des Mousses Port-Cartier, QC G0W 1X0, Canada 50.025323, -66.818910 Charm Industrial Miniforge 2360 S Rollie Ave Fort Lupton, CO, USA 80621 40.076570, -104.804680 Kerry, Inc. Red Arrow 4343 Expo Drive Manitowoc, WI 54220 44.080613, -87.704757 <u>Bio-oil Injection & Storage:</u> Charm Industrial (Basco Injection site)

	Evangeline Parish, LA 30.916251, -92.297908
Verification Type	Isometric Standard credit verification with project expansion in accordance with the following standard, protocol, and modules: - Isometric Standard V1.8 - Bio-oil Geological Storage Protocol v1.2 (Kerry Bio-oil), v1.1 (AECN and Charm Bio-oil) - Biomass Feedstock Accounting Module v1.3 (Kerry Feedstock), v1.2 (AECN and Charm feedstocks)* - Biomass or Bio-oil Storage in Permeable Reservoirs v1.2 - Embodied Emissions Accounting V1.0 (AECN and Charm Bio-oil) - Energy Use Accounting V1.2 (AECN and Charm Bio-oil), V1.3 (Kerry Feedstock) - Transportation Emissions Accounting V1.2 (AECN and Charm Bio-oil) - GHG Accounting V1.1 (Kerry Bio-oil)

*Note that Charm Bio-oil produced after December 2025 utilizes new feedstock validated under Biomass Feedstock Accounting Module V1.3

2. Technology Description

Charm obtained bio-oil from three sources during this reporting period. Charm obtains bio-oil from a producer in Quebec, Canada (AECN) who uses an Ensyn/Honeywell UOP pyrolysis system to generate bio-oil from waste wood produced at a neighboring mill.

In addition, Charm produces bio-oil at its pyrolysis facility in Fort Lupton, CO. Production of bio-oil for sequestration and carbon removal credit generation began in August of 2024, with the initial installation of four pyrolyzers, with plans for up to 18 pyrolyzers total to be installed. Charm's pyrolyzers utilize wood chips, placed into a high temperature pressurized reactor in a low oxygen environment to reduce the wood chips into a combination of products, including syngas and light hydrocarbon gases, pyrolysis oils, biochar, and wood vinegar. All hot pyrolysis gases from the reactor are quenched with water to produce the oil and wood vinegar products. Excess gases are routed to a combustion device for emissions control, with the heat produced from the combustion of excess gas utilized for drying the biomass from 40%-50% moisture content to 10% moisture content before introduction into the pyrolyzer.

Finally, Kerry Inc pyrolyzes woody biomass to produce liquid smoke, a popular flavoring product. This process produces a carbon-rich liquid pyrolysis oil which is intentionally phase separated. The "light ends" of the separated product are refined to produce the primary product of the system. The "heavy ends" are a waste product with high biogenic carbon content. While some of that product is combusted on-site to generate energy, the remainder is typically disposed of by incineration without energy capture. Charm has contracted with Kerry to purchase this waste material for durable carbon removals.

Bio-oil from AECN, Kerry, and Charm, and aqueous fraction product from Charm pyrolyzers are shipped by tanker truck, rail car, or tote tanks on flatbed to the injection facility in Louisiana. At this facility, Charm pretreats the bio-oil via air sparging and any other required adjustments to ensure bio-oil and aqueous product meet the requirements of the injection facility permit for storage in permeable reservoirs as specified by the Louisiana Department of Energy & Natural Resources (LDENR).

The bio-oil products are blended in on-site storage tanks through recirculating pumps and injected as soon as practicable. Samples of the blended bio-oil are collected prior to injection and analyzed for bio-oil characteristics to ensure compliance with permit requirements. Charm also monitors all injection parameters as required by the Underground Injection Control (UIC) permit.

2.1. Changes since Validation and Previous Verification

Several minor changes, updates, or unique scenarios were identified by Charm for the current reporting period regarding the Charm Industrial processes, data collection, and calculation procedures which have been specified in the GHG statement and supporting documentation. The following modifications and notes have been provided for clarity. All were reviewed and deemed acceptable or non-impactful under the Isometric requirements:

- For AECN rail offloads into trucks, net bio-oil mass injected at the storage site is the Full scale weight minus the Empty scale weight of the same tractor + trailer. When an offload's empty scale ticket is missing, or its trailer/tractor does not match the Full ticket (a rig mismatch), the recorded empty weight cannot be used directly. In these cases, a conservative per-trailer standard empty weight is substituted.
- In this Reporting Period, Kerry oil is delivered to the Basco injection site in bulk tanker trailers rather than in IBC totes as in prior Periods. Delivered mass is therefore established from CAT Scale tickets — the full weight less the post-offload empty (tare) weight of the same tractor and trailer — rather than from per-tote tag weights.
- Kerry oil passes through the same wellhead filter as Charm's other injected oil, and the solids retained are subtracted from the delivered mass to arrive at the mass of bio-oil injected. For Kerry, that subtraction uses the measured solids content of Kerry calciner oil itself, assuming all solids are removed and deducted from the delivered mass of bio-oil injected.
- For this reporting period and going forward, Charm has moved to a process of calculating the carbon intensity of the Charm oil and other pyrolysis products on a 6-month basis. The data from the prior six months of production is used to compute the carbon intensity (CI) of each product in the same way it is computed in monthly Production Periods. For the subsequent six months, those CIs are used to compute production emissions netted against removals for all Charm bio-oil. In this case, the data from January 1 – June 30, 2026 (H1 2026) sets the CI which is in turn used for removals injected July 1 – December 31, 2026.

3. Verification Process

3.1. Objectives and Scope

A planned series of verification activities were conducted by 350Solutions to independently verify the GHG Statement submitted by Charm Industrial for Charm Bio-oil Geologic Storage following the guidelines of the criteria listed in Section 3.5. The intention of the verification is to confirm GHG removals in accordance with the validated Project Design Document (PDD).

3.2. Responsibilities

The project proponent, Charm Industrial, is responsible for the fair presentation of the GHG Statement in accordance with Isometric criteria.

It is the responsibility of 350Solutions to express an opinion as to whether the GHG Statement and supporting documents support verification under the relevant Isometric Standard, protocol, and modules.

3.3. Level of Assurance & Materiality

This verification was completed with a reasonable level of assurance. The materiality threshold of 5%, considering the totality of all omissions, errors, and misstatements, was applied to this audit.

3.4. Validation and Verification Body Details

350Solutions Inc. declares that we are an impartial verifying body, free from any conflicts of interest, capable, and qualified to complete this verification for the current operational period according to the Isometric Standard V1.8 and applicable protocols and modules utilizing a reasonable level of assurance.

350Solutions is an accredited inspection & verification body by ANAB under ISO 17020 for completion of ISO 14034 Technology Verifications. 350Solutions is based out of Raleigh, North Carolina, USA. The Technical Lead for the Charm Bio-oil Geologic Storage verification is Tim HansenQuality assurance was provided by Lily Schacht. Complete verifier qualifications are included in Appendix 3.

3.5. Protocols and Modules Applied & Applicability

The verification was conducted following the specifications of the following standards, protocols, and modules:

- Isometric Standard V1.8
- Bio-oil Geological Storage Protocol v1.2 (Kerry Bio-oil), v1.1 (AECN and Charm Bio-oil)
- Biomass Feedstock Accounting Module v1.3 (Kerry Feedstock), v1.2 (AECN and Charm feedstocks)*
- Biomass or Bio-oil Storage in Permeable Reservoirs v1.2
- Embodied Emissions Accounting V1.0 (AECN and Charm Bio-oil)
- Energy Use Accounting V1.2 (AECN and Charm Bio-oil), V1.3 (Kerry Feedstock)
- Transportation Emissions Accounting V1.2 (AECN and Charm Bio-oil)

- GHG Accounting V1.1 (Kerry Bio-oil)

Bio-oil supplied by AECN and Charm bio-oil produced prior to December 2025 were validated under Biomass Feedstock Accounting Module v1.2 and Bio-oil Geologic Storage Protocol v1.1 and remain validated under these modules. Bio-oil provided by Kerry, Inc. and Charm bio-oil produced in December 2025 and later are validated under updated versions of the feedstock module (v1.3) and storage protocol (v1.2).

Charm utilizes Book and Claim Units (BCUs) to substitute for some, or all, of project diesel fuel usage and purchases Renewable Energy Certificates (RECs) for some, or all, of its electricity consumption in accordance with the Isometric Energy Use and Transportation Modules.

The Charm project takes mixed woody waste or forestry thinnings from wildfire prevention activities, converts it to bio-oil, which they pretreat and inject into geologic storage formations. The biomass feedstocks were assessed previously under the eligibility criteria of the Isometric Biomass Feedstock Module and were found to be eligible. As the biomass feedstocks have been validated as eligible biomass feedstocks, the storage of biomass conversion products (bio-oil) in a geologic formation makes both the Bio-oil Geological Storage Protocol and Biomass or Bio-oil Storage in Permeable Reservoirs Module applicable to this project. All other modules for energy, transportation, and embodied emissions accounting are generally applicable to this project.

3.6. Project Boundary

The boundaries of the project include the supply of wood waste to the biomass processing site, the conversion of the biomass to bio-oil via Charm, Kerry, or AECN's pyrolysis process, transport of the bio-oil to Charm's injection site, completion of pretreatment, and injection of the bio-oil in the geologic reservoir. Monitoring and required analysis of the bio-oil and the storage formation are also included in the project boundary as are all embodied emissions, including end-of-life stage, associated with equipment and consumables.

3.7. Approach

The verification process activities are described in Table 2. Appendix 1 contains findings identified during the verification process and their resolutions. The files that were reviewed are listed in Appendix 2. Audit results are documented using 350Solutions' Compliance Checklist, which includes all requirements from the Isometric standard, Bio-oil Geologic Storage protocol, and referenced modules.

Table 2. Verification Activities.

Date(s)	Activity	Tasks
August 12-13, 2026	Evidence Review and Verification	- Review validated PDD - Verification of claims made in GHG statement - Verification of associated claims (social and environmental safeguarding) - Review supporting documentation - Review of prior Verification Report and findings

		- Identification of major and minor process steps, measurements, and calculations and changes in the reporting period - Review GHG Statement entries and evidence on Isometric Certify Platform and issue Log of Issues - Review supplier responses to Log of Issues and close out issues as possible
August 13-14, 2026	Report Writing	- Draft report - Internal Quality Assurance - External review

4. Crediting Details

4.1. Project Reporting Period

This verification considers injections of bio-oil (oil and aqueous phases) by Charm during the period July 1 – 31, 2026. The reporting period is based on the actual injection dates, with bio-oil production, transportation, monitoring, and related activities potentially occurring outside of this reporting period but properly accounted for and allocated to the bio-oil injected within the reporting period.

4.2. Crediting Period

The 5-year crediting period for the Charm Bio-oil Geologic Sequestration Project is March 1, 2024 – February 28, 2029. All claimed removals in this reporting period fall within the current crediting period.

5. Verification Summary

5.1. Accuracy of the Opinion

Confirmation of the GHG statement has been based on review and confirmation of calculations of total bio-oil and CO_2 equivalent injected and all process, operational, and embodied emissions, as well as all supporting evidence for the following removals listed in Table 3, representing two of the three weeks of injections and all process emissions. The operations sheet for the remaining removals (from Week 3) was compared to the Certify inputs to check for errors during data transfer.

Table 3. Removals Reviewed in Detail

Removal ID	Description
rmv_1KZBJBN9C1S0M5R5	AECN Rail delivery Injection
rmv_1KYT7WF6Q1S03R7Q	AECN Rail delivery Injection
rmv_1KYZ3JPB51S0VE5Q	Kerry Bio-oil injection
rmv_1KYWCXFBK1S0Y3NV	Charm Aqueous Injection
rmv_1KYWMYBJ51S0HH3N	Charm Bio-oil Injection

For the removals listed in Table 3, the review included:

- Re-calculation of mass of bio-oil delivered and injected including tracing to raw data (weigh scale tickets, carbon content lab results, addition of consumables)
- Confirmation of transportation distances for tanker and rail car deliveries from AECN, Charm, and Kerry to Basco site, LA, emission factors, and calculation of transportation emissions
- Confirmation of weigh scale raw data for deliveries of bio-oil and injectate
- Confirmation of Charm oil production emissions and carbon intensity for H1 2026
- Confirmation of raw data for and recalculation of total fuel usage at Charm injection facilities
- Confirmation and recalculation of total electricity usage at Charm injection facilities
- Confirmation and recalculation of proper use and accounting of BCUs and RECs to offset diesel and electricity usage emissions, respectively
- Confirmation and verification of updated embodied emissions totals for added equipment during the reporting period and integration in production period carbon intensity
- Review of Basco injection site emissions and supporting evidence (fuel and electricity use)

Additionally, the review included the Basco site injection (monitoring data and permit reporting requirements) and post-injection monitoring to ensure compliance with Class V permit for Basco site and the updated stakeholder engagement activities and grievance log.

The verified totals across all removals are listed in Table 4.

Table 4. Summary of Removals

Parameter	tCO_2e
Net	874.29
Sequestrations	1,001.88
Activities	-197.11
Removal counterfactuals	0
Losses	0
Adjustments (Book and Claim Units)	75.00
Uncertainty discount⁺	0

* Values have been rounded and may not add up without error.

⁺ Charm incorporates uncertainty in measurements within the data and calculations reviewed.

5.2. Verification Opinion

Based on the confirmation described in Section 5.1, 350Solutions has found that the data presented in the Charm Bio-oil Geologic Storage GHG Statement and supporting documentation:

☒ ***Meets the requirements of Isometric Standard V1.8 and Bio-oil Geological Storage Protocol*** v1.2 (Kerry Bio-oil), v1.1 (AECN and Charm Bio-oil) ***for Verification***

☐ ***Meets the requirements of Isometric Standard V1.8 and Bio-oil Geological Storage Protocol*** v1.2 (Kerry Bio-oil), v1.1 (AECN and Charm Bio-oil) ***for Verification with minor modifications***

☐ ***Does Not Meet the requirements of Isometric Standard V1.8 and Bio-oil Geological Storage Protocol*** v1.2 (Kerry Bio-oil), v1.1 (AECN and Charm Bio-oil) ***for Verification***

By adhering to the requirements of the Isometric standard, protocols and modules, removals of 874.29 tonnes CO2e from Charm Industrial's Charm Bio-oil Geologic Storage have been verified in accordance with the validated PDD and the criteria listed in Section 3.5 with a reasonable level of assurance.

VVB	Lead Auditor	Report ID No.
350Solutions, Inc.	Tim Hansen	350-IS.CHA.BO.18-VR

Signature: Timothy Hansen (Aug 14, 2026 16:20:22 EDT)
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Signed: Tim Hansen (Lead Auditor) Lily Schacht (Quality Assurance)

5.3. Summary of Verification Findings

Table 5 contains a summary of findings and any identified issues associated with each requirement in the applicable Isometric standard, protocol, and modules. Assessment of the greenhouse gas removal statement identified no non-conformities, with only a few minor findings, clarifications, and non-material data issues (see Appendix 1). No significant findings that materially impacted the verified net carbon removals nor compliance with protocol standard were identified.

Table 5. Requirements from the relevant Isometric Standard, protocol, and modules.

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
2.1.	*Validated against approved Protocols*	N	Acceptable – No additional changes since validation
2.5.1	*Boundaries*	N	Acceptable – No additional changes since validation
2.5.2	*Baseline*	N	Acceptable – No change since validation
2.5.3	*Additionality - Financial*	N	Acceptable – No change since validation
2.5.3	*Additionality – Environmental*	N	Acceptable – No change since validation
2.5.3	*Additionality – Regulatory*	N	Acceptable – No change since validation
2.5.4	*Leakage*	N	Acceptable – No change since validation and prior verification.
2.5.5	*Default Emission Factors, Proxies, and Models*	N	Acceptable – No additional significant changes were made since prior verifications.
2.5.11	*Life Cycle Assessment (LCA) Policies*	N	
2.5.6	*Common Calculation Factors*	N	
2.5.7	*Uncertainty*	N	
2.5.8	*Durability and Monitoring*	Y	All updated durability evidence and storage monitoring requirements validated to meet protocol requirements. Note that ongoing discussions with Isometric and Charm are occurring regarding frequency of Kansas storage site monitoring reports as the prior injections transition into a post-injection / closure monitoring phase with no Charm injections occurring. Full verification of monitoring for existing storage at Kansas site anticipated in August, 2026
2.5.9	*Risk of Reversal*	N	Acceptable – No additional changes since validation

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
2.5.10 3.7	*Environmental and Socioeconomic Impacts*	N	Acceptable – No additional changes since validation.
3.1	*Ownership*	N	Acceptable – No change since validation
3.2	*Documentation*	Y	Acceptable – No additional changes since prior verification. All documentation properly updated for reporting period
3.3	*Eligibility*	Y (GHG Statement Updated)	Acceptable – Project registered with Isometric and reporting period GHG Statement properly submitted to Isometric dMRV Certify platform.
3.4	*Project Crediting*	Y (GHG Statement Updated)	Acceptable – New reporting period GHG Statement is complete and verified. Reporting period is within current validation crediting period.
3.5	*Stakeholder Input Process*	N	Acceptable – No changes since prior verification. Community Engagement & Grievance Records reviewed.
3.6	*Regulatory Compliance*	Y (Required Monitoring Updated)	Required monitoring and reporting completed during reporting period. No additional compliance issues identified for injections during period.
Biomass Feedstock Accounting Module			
2.1	*Eligibility Criteria for Market Leakage*	N	Acceptable – No change since prior feedstock validations
2.2	*Counterfactual Storage Eligibility*	N	Acceptable – No change since prior feedstock validations
2.3	*Dedicated Energy Feedstock Eligibility*	N	Acceptable – No change since prior feedstock validations
3.0	*Counterfactual Definition*	N	Acceptable – No change since prior feedstock validations
3.1	*Counterfactual Energy Definition*	N	Acceptable – No change since prior feedstock validations
3.2	*Counterfactual Replacement Definition*	N	Acceptable – No change since prior feedstock validations
Bio-oil Storage in Permeable Reservoirs Module			
1.0	*Storage System Eligibility*	N	Acceptable – No change since validation
2.1	*Storage Reservoir – Permitting*	N	Acceptable – No change since validation
2.2	*Storage Reservoir – Capacity*	N	Acceptable – No change since validation
2.2	*Storage Reservoir – Integrity*	N	Acceptable – No change since validation
2.2	*Storage Reservoir – Integrity: Characterization*	N	Acceptable – No change since validation.

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
2.2	*Storage Reservoir Characterization – Temperature, pH, conductivity, density*	N	Acceptable – No change since validation
2.2	*Storage Reservoir Characterization – $d^{13}C$ analysis (where applicable)*	NA	Acceptable – Not required
2.3	*Storage Reservoir Integrity – Well Construction*	N	Acceptable – No change since validation
3.1.1	*Storage Reservoir Integrity – Injection operations*	Y	Acceptable – Updated monitoring data provided indicating no exceedances of permit limits for Basco site
3.1.1	*Storage Reservoir Integrity – Injection operations – Sampling & Analysis*	Y	Acceptable – Updated sampling and analytical data provided indicating no exceedances of permit limits for Basco site
1.0, 3.2	*Well Closure and Post Closure Requirements*	N	Acceptable – No change since validation
3.1.3	*Injection Pressure*	Y	Acceptable – Pressure is continuously monitored and reported quarterly (monitoring reports indicate full compliance) for Basco site
3.1.2	*Mechanical Integrity*	N	Acceptable – No change since prior verification for Basco site
3.1.3	*Volatile Gas Monitoring*	N	Acceptable – No change to processes since prior verifications for Basco site.
3.1.3	*Reversal Risk & Monitoring*	N	Acceptable – No change since validation
3.3.1	*Attribution of Reversals*	N	Acceptable – No change since validation

5.4. Forward Action Requests & Recommendations

Forward action requests (FAR) and recommendations are also provided in Appendix 1 for consideration for improvement of future operations and GHG Removal Credit calculations and to ensure tracking of potential issues to be monitored by verifiers in future verifications. Open FARs and opportunities for improvement are summarized in Table 6, below, for future verification reference.

Table 6. Forward Action Requests

ID No	Type	Finding / Issue	Supplier Response	Conclusion / Resolution
NA	FAR (prior)	With injections no longer occurring at Vaulted, we would like to know how Charm will be notified of closure of the cavern where Charm oil was last injected, and to ensure that Charm will now be obtaining and submitting any post closure monitoring data to Isometric as part of the continued monitoring requirements. Please clarify that plans are in place for this and will be ongoing.	Charm has reached out to Vaulted on multiple occasions to get clarity regarding their future plans for the AR-09 injection well. The well still has considerable capacity and is the property of Vaulted, so it is up to them whether they plug and abandon the well or use it for their own emplacements. This decision on their part will affect the nature of the post-closure monitoring plan. We will continue to request clarity regarding these plans and, once received, we will define and submit a monitoring plan in light of the information we receive.	April 2026: guidance provided by Isometric for review and implementation with verification of monitoring proposed in upcoming RPs. May 2026: Anticipate compliance verification to occur in June or July RP and FAR to be closed then. June 2026: Confirmed compliance will occur during the July RP July 2026: Compliance data has been collected in July and will be submitted for monitoring verification in August 2026
NA	FAR (prior)	With injections no longer occurring at Vaulted, please provide updated contract for post-emplacement / post-closure management and monitoring and post-closure monitoring plan that documents how Charm will be notified of closure of the cavern where Charm oil was last injected, and to ensure that Charm will now be obtaining and submitting any post closure monitoring data to Isometric as part of the continued monitoring requirements or this storage location		See above
ISS-36	FAR (prior)	The Basco site emissions allocation approach will result in a significant back-end emissions allocation based on a massive scaleup of the Charm oil production and storage. This approach should be		Based on discussion with Charm: Charm has provided significant additional documentation (containing proprietary forecasts) documenting

		reviewed annually, along with progress toward forecast injection totals from the three feedstocks, to ensure embodied emissions allocations are allocated properly and not misallocated. Can Charm please clarify the Charm bio-oil scaleup approach and the progress, considering the Charm production is lagging significantly behind the forecast used for the Charm production period embodied emissions? In addition - it doesn't appear that the forecasts used here for Charm oil injection match the forecasts used of Charm production in the Charm embodied emission calcs (off by an order of magnitude for out years?)		approach for reconciling this and accounting for reconciliation of embodied emissions allocations. Moving to forward action request based on evidence provided of forward actions, with discussion and approval with Isometric, Charm and 350 to occur. July 2026: Updated amortization plan proposed and under review. To be implemented from August 2026 RP forward.
ISS-34	FAR (prior)	In reviewing the Nov 2025 PP embodied emissions allocations, it was observed that the embodied emissions allocations are lagging significantly behind due to production levels not hitting forecasts. When reviewing Cap Ex MRV November spreadsheet, the forecast biomass throughput for the production periods completed to date (through Nov 2025) are forecast to have a total biomass throughput of 2749 tDBM, when the actual throughput has been 1019 tDBM, or 37% of forecast. As a result, the embodied emissions will not be fully accounted for in the 5 year allocation period planned without future production significantly exceeding		See above.

		planned capacity. It is suggested that the allocation schedule be revised to reflect actual operations and revised production forecasts (noting max throughput has been 116 (average 81) tonnes per month as opposed to forecast 234 tonnes). This will significantly increase embodied emissions allocation for all periods, (close to double) and would require accounting for prior production periods as well.		

6. Revision History

Version	Date Issued	Noted Changes
Draft Version 1.0 (V1.0)	August 13, 2026	N/A
Version 1.1 (V1.1)		Minor edits following internal QA
Version 1.2 (V1.2)		Minor edits following external review

Appendix 1: Log of Findings

Issue Code	Date Issued	Type	Title	Body	Status	Responses	Resolution	Resolved At
ISS-86	13 Aug 2026	Clarification Request	Thermal Oxidizer Temperature Excursions	The Thermal Oxidizer temperature monitoring trace (July CatOx -- Processed.xlsx) shows several excursions below the 600F minimum setpoint. Please clarify the issues with reduced temperatures (i.e. if these are startup or shutdown of oxidizer periods, if these are during times when no sparge gas is being fed, etc.) and clarify the compliance status with the permit requirement	Resolved	Charm Industrial User 13 Aug 2026: "These excursions generally happen for two reasons 1. The oxidizer is operating normally, but VOCs are not being pushed out of the oil and over the catalyst. That means that air is being blown over the inlet without triggering an exothermic reaction. This can cool the inlet temporarily, but by definition also means that VOCs are not being released through the blower stream 2. The oxidizer has been shut down -- This is often triggered by a runaway exothermic reaction or water from the knockout drum being pulled into the oxidizer. In both cases, an emergency stop is triggered, which causes the unit temperature to fall. However, when this happens the blower is cut off as well, as discussed in the oxidizer brief provided with May reporting documentation. This acts as an engineering control which prevents sparged	Explanation provided. Since the two reasons either mean there is a high or a low VOC concentration, can you add an indicator to the graph to show if the blower is off, which would allow us to distinguish between reason 1 and reason 2?	14 Aug 2026

						VOCs from being pushed out of the tank until the unit has been put back in service. Regarding permit compliance risk: The permit is for 4.78 tons of VOCs, which is based on the total "potential to emit" that would result from the unabated volatization of all VOCs in the bio-oil we expect to process in a year. In sum: The excursions you have noticed don't differ from the expected behavior of the unit, and the blower control prevents uncontrolled emissions in the case of an e-stop. Beyond that, even if there were a transitory uncontrolled emission, it would not exceed the permit threshold. "		
ISS-85	*12 Aug 2026*	*Omission*	*Upload Mass evidence*	*For additional traceability, please upload the waste manifest with mass measurement evidence to Certify unless there is a confidentiality reason not to (noting it is provided in Google Drive).*	*Resolved*	*Charm Industrial User 13 Aug 2026: "File has been attached. Needed to adjust file type to jpg in order to be accepted, so let us know if there is any corruption and we can readjust. "*	*The converted file isn't visible. Acceptable for now considering it was provided in Google Drive, but please adjust for future reporting periods. *	*14 Aug 2026*
ISS-84	*12 Aug 2026*	*Clarification Request*	*No. of weeks / mo is not always 4*	*Just noting that the multiplier of weeks in the calculation sheet is currently 4. For July, it is a 4.5 week month. There should be 4 months per year where 5 weeks is used as the multiplier to be accurate. Noting this is non-material.*	*Resolved*	*Charm Industrial User 13 Aug 2026: "The point is well taken. Using days rather than weeks to scale this value should address this issue. We will use that approach going forward."*	*Will be adjusted in future reporting periods for more accurate emissions estimation*	*14 Aug 2026*

ISS-83	*12 Aug 2026*	*Omission*	*Evidence for distance from carrier?*	*The tab in the evidence file indicates a list of distances provided by the carrier. In other submissions, an email or other record is provided from the carrier. Is similar evidence available for this value (552 miles)?*	*Resolved*	*Charm Industrial User 13 Aug 2026: "A screenshot of an email from the carrier confirming these distances is now attached to the component."*	*Email evidence provided*	*14 Aug 2026*
ISS-82	*12 Aug 2026*	*Clarification Request*	*Rail Mileage mismatch*	*Can you please explain why the total rail mileage (Cells K10-13 of AECN Rail tab in Ops Notes) totals 8796 and the value utilized cell M48 - a sum of allocated rail mileage to tankers and injected loads (8779.05) are different? Shouldn't they be the same? It is a non-material difference - just trying to understand the math and approach.*	*Resolved*	*Charm Industrial User 13 Aug 2026: "This is a result of small drift from rounding and significant figures in this mass based distance derivation. When full railcars are continuously injected like in this week, Charm agrees that we should use total distances because of this potential drift in larger injection weeks. (No mass based derivation is required.) Going forward, this mass correction will be implemented so we do not encounter this drift."*	*Explanation provided - small rounding error*	*14 Aug 2026*
ISS-81	*12 Aug 2026*	*Omission*	*Map attached is a mismatch*	*The attached map in certify only shows one apparent leg of the trip (173km). Please upload additional image that demonstrates the total trip distance*	*Resolved*	*Charm Industrial User 13 Aug 2026: "The value for this component is an aggregate of multiple trips at the attached distance. The math for how this was derived is L48 in the Ops Notes."*	*Clarification provided*	*14 Aug 2026*
ISS-80	*12 Aug 2026*	*Clarification Request*	*AECN CI timing*	*Can the AECN CI be updated any more quickly than currently? It appears that the H2 2025 value is still being used although it is currently H2 2026. Would prefer*	*Resolved*	*Charm Industrial User 13 Aug 2026: "Happy to request that. We've just received the CI for H2 of this year, so that will be*	*Charm will request more timely CI updates from AECN*	*14 Aug 2026*

				*updating within 30 days of the half year if possible? *		*used in reporting going forward."*		

Appendix 2: Documents Submitted for Review

Table A4.1. Documents submitted on Isometric's Certify Platform

Appendix I: Reference Table

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src_1KZ1T7YEC1S08199	2382-R.pdf
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src_1KYZ4RYHT1S0ZA4T	W4 Kerry-AECN SGS COA.pdf
src_1KYZ4S0Q21S0JZ0F	W4 Kerry CAT Scale Ticket 4921 FULL.jpg
src_1KYZ4S4VA1S06KW4	W4 Kerry CAT Scale Ticket 8165 EMPTY.jpg

All sources referenced in the GHG Statement (capped at first 100 of 164 sources):

Table A4.2. Documents submitted through Google Drive

Pending

Appendix 3: Verifier Qualifications

Lily Schacht
Director of Science, 350Solutions

EDUCATION:
MS, Environmental Engineering, University of Wisconsin – Madison, 2019
BS, Chemical, Energy, and Environmental Engineering, Washington University in St. Louis, 2017

EXPERIENCE SUMMARY:
Lily Schacht is an Environmental and Chemical Engineer with experience in process engineering, environmental chemistry, analytical methods, and life cycle analysis (LCA). At 350Solutions, Lily works on verifying carbon dioxide removal (CDR) technologies, with a focus on mineralization-based pathways, including enhanced weathering, direct air capture, and ocean alkalinity enhancement. Previously, Lily led agronomic research at an enhanced weathering CDR supplier where she organized field trials across multiple states to quantify carbon removal and crop yield changes after rock application. Before that, Lily developed rapid prototyping instrumentation to optimize a biomineralization process in concrete production and aided in scaling up the process to pilot-scale. Lily also built environmental impact models for process variable sensitivity analysis of demo-scale manufacturing processes. These models were used to drive the direction of research and development to minimize product life cycle impacts. Throughout her career, Lily has developed analytical chemistry methodologies for both liquid- and solid-phase analyses.

RESEARCH AND PROFESSIONAL EXPERIENCE:
March 2024 – Present: Director of Science, 350Solutions
Verify CDR technologies on behalf of registries and the XPRIZE Carbon Removal challenge. Specializes in mineralization pathways, including mineralization kinetics, measurement methods, and open-system modeling.
Nov 2022 – Feb 2024: Researcher, Lithos Carbon
Quantified carbon removal rates and agronomic impacts of enhanced weathering on cropland across six US states; Evaluated chemical analysis methods for precision relative to cost.
Nov 2023 – Jan 2024: Independent Consultant, Keel Labs
Built an environmental impact model to evaluate potential material and process changes; Recommended areas for reducing material usage up to 80%
Dec 2020 – Oct 2022: Research Scientist II, Biomason
Guided experimentation and data analysis throughout all R&D teams to inform techno-economic analysis (TEA) and LCA; built and led the carbonate biomineralization prototyping workstream for rapid iteration; developed real-time measurement techniques for critical process parameters in solid state

SELECTED PUBLICATIONS & PRESENTATIONS:

- **Schacht, L.**, Baum, M., Liu, H., & Yap, M. (2023) Scaling Enhanced Rock Weathering: Agronomic Impacts at Field-Scale [Abstract]. ASA, CSSA, SSSA International Annual Meeting, St. Louis, MO.
- **Schacht, L.** and Ginder-Vogel, M. Arsenite Depletion by Manganese Oxides: A Case Study on the Limitations of Observed First Order Rate Constants. *Soil Syst.* 2018, 2(3), 39. https://doi.org/10.3390/soilsystems2030039

Tim Hansen, P.E.
Founder and CEO, 350Solutions

EDUCATION:
B.S., Chemical Engineering, University of Virginia, 1993
M.S., Engineering Science, Thayer School of Engineering, Dartmouth College, 1995

EXPERIENCE SUMMARY:
Mr. Hansen has 26 years of experience in management of energy and environmental technology development and demonstration projects and programs, as well as multimedia environmental engineering efforts. The majority of his recent work has focused on the evaluation of innovative carbon capture, utilization, and removal technologies. Mr. Hansen has led the development and management of large technology evaluation programs in the advanced energy, transportation, and climate change areas.

RESEARCH AND PROFESSIONAL EXPERIENCE:
2019-Present Founder – CEO, 350Solutions, Inc.
Owns and operates a small cleantech engineering consulting business focused on the independent evaluation of new cleantech innovations and their impact on the environment and carbon emissions. Provides engineering consulting, testing and evaluation, techno-economic assessment, and other support to companies developing, using, or investing in new clean technology innovations. Manages administrative, business development, and project activities for 350Solutions.

2012-2019: Director - Energy and Environment, Southern Research
Manages scientific and technical staff performing research, development, and evaluation of innovative clean energy technologies. Projects range from $25,000 to $6million in size, and are funded by the US Department of Energy, Department of Defense, and commercial partners. Technical focus areas are conversion of biomass to fuels and chemicals, carbon capture and utilization, energy efficient building technologies and renewable energy generation.

2009-2012: Program Manager – Transportation & Climate Change Technology, Southern Research
2003-2009 Sr. Project Leader, Environmental Engineer, Southern Research
1996-2003 Environmental Engineer, Bensinger & Garrison Environmental

PROJECT EXPERIENCE:
Mr. Hansen has executed several independent technology performance verifications of emerging carbon, energy and transportation technologies, as CEO of 350Solutions, Director of Energy & Environment at Southern Research, and Director of the U.S. EPA's Greenhouse Gas Technology Center. Mr. Hansen has completed clean technology evaluations for the Department of Defense, state energy agencies, commercial clients, investors, and technology developers, involving evaluation of commercial feasibility, economic and environmental impacts, and technology performance. Mr. Hansen served as the Measurement and Verification Program Lead for the NRG COSIA Carbon XPrize – a $20M prize competition for technologies that capture and beneficially utilize CO_2. Mr. Hansen also served as U.S. Technical Expert for the development and implementation of ISO 14034 – Environmental Technology Verification, an international standard, issued in 2016

Verification Report

Final Audit Report 2026-08-14

Created:	2026-08-14
By:	Timothy Hansen (tim@350Solutions.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAULZ24p2QofHa8lTkJtoiMPl6jD0Npm2i

"Verification Report" History

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